

Srikar Pothapragada

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PROFESSIONAL EXPERIENCE

Data Engineer

August 2024 – Present

O'Neill School of Public and Environmental Affairs

Remote

- Built a county wise healthcare data analysis framework using PySpark and MySQL, analyzing 50M+ healthcare records from NCHS to identify high-need areas and guide donor fund allocation with data driven insights using Tableau.
- Fine-tuned SpaCy, BART, and RoBERTa to develop a text classification pipeline, categorizing 300K+ philanthropic mission statements by NTEE codes and keywords, enhancing donor targeting and fund allocation efficiency.
- Collaborated to build a RAG chatbot using FAISS vector search, Mistral LLM, and Text-to-SQL to convert natural language queries into SQL, integrated question reframing and query caching for better accuracy and reuse.

Cloud Machine Learning Engineer

May 2024 – August 2024

Indiana University

Bloomington, IN

- Implemented an end-to-end, serverless pipeline using AWS S3 event triggers, Step Functions, and Lambda with Dockerized Whisper, automatically transcribing 100+ hours of audio into timestamped SRT files on upload.
- Enhanced transcript alignment by deploying Stanza NLP and n-gram mapping in AWS Fargate, automating long-running, resource-intensive batch processing and reducing timing mismatches by 60%.
- Collaborated with linguists to automate merging of acoustic and phonological features, generating vowel space and duration plots with improved speaker variation insight.

Software Engineer

July 2021 – July 2023

LTIMindtree

Hyderabad, India

- Engineered scalable ETL pipelines using Airflow to automate daily ingestion of 10M+ records into S3 via AWS DMS and Glue, leveraged PySpark for distributed transformations, ensuring high-throughput, fault-tolerant data processing.
- Designed Delta Lakes on S3 using PySpark (EMR) to enable ACID transactions, schema enforcement, and time travel.
- Enabled real-time Change Data Capture (CDC) using AWS Glue and engineered SCD Type 1/2 logic for historical tracking; modeled Star Schemas and leveraged Redshift Spectrum for high-performance, federated analytics.
- Improved Redshift query performance by 50% by optimizing SQL, creating materialized views, and using stored procedures.
- Enforced data governance by configuring IAM roles, S3 policies, and KMS encryption for securing PII/PHI.
- Enabled CI/CD for AWS ETL pipelines using Terraform and GitHub Actions, ensuring reliable, zero-downtime deployments.
- Mentored 3 interns to build Spark SQL pipelines that deduplicated data and flagged anomalies, boosting data integrity.

TECHNICAL SKILLS & CERTIFICATIONS

Programming Languages : Python, R, SQL/T-SQL, JavaScript, TypeScript, C, C++

Database and Analytics : MySQL, Microsoft SQL Server, PostgreSQL, Snowflake, MongoDB (NoSQL), Power BI, Tableau

Web Frameworks & API Development: React, Node.js, React Native, FastAPI, REST APIs

Big Data & DevOps Tools : Hadoop, Apache Spark, Apache Airflow, Apache Kafka, Docker, Kubernetes, GIT, CI/CD

Machine Learning & Deep Learning : Scikit-learn, PyTorch, TensorFlow, NLTK, SpaCy, Whisper, Praat, RAG, CNN, RNN

Cloud Technologies : AWS (S3, Glue, Redshift, EMR, Kinesis, RDS, DynamoDB, Athena), Azure, GCP.

Certifications : AWS Data Engineer Associate, Oracle AI Vector Search Certified Professional, Microsoft Fabric Data Engineer Associate

PROJECTS

CrimsonAI Multi-Agent RAG Application | [GitHub](#) | *LangChain, Claude, pgvector, RAG, FastAPI.*

- Architected a multi-agent RAG system that retrieves personalized course recommendations through vector-based semantic search, implemented with FastAPI and React/Redux for an intuitive interface enabling contextual academic guidance.

StreamLineYT | [GitHub](#) | *Apache Kafka, Spark, Airflow, MongoDB, Docker.*

- Developed a real-time YouTube analytics pipeline to stream video metadata and perform scalable topic modeling using LDA and BERTopic to uncover content trends and improve insight into audience engagement patterns.

Cross-Platform News Aggregator App | [GitHub](#) | *React Native, Expo, RESTful APIs, Appwrite*

- Designed a cross-platform mobile app using React Native and Expo, integrating Appwrite for secure auth and News API for real-time content, with features including category-based filtering, trending news, user preferences, and dark mode UI.

EDUCATION

Indiana University Bloomington (GPA: 4.0/4.0)

August 2023 – May 2025

Master of Science in Data Science

Bloomington, IN

Relevant CS coursework - Data Structures and Algorithms, Distributed Systems, Advanced Database Concepts, Mobile App development, Operating Systems, Computer Networks, Object-oriented Programming.